



Investigation of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) Perovskite Thin Films Produced by Ultrasonic Spray Pyrolysis Method

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Abstract

Methylammonium iodide (MAI) and methylammonium bromide (MABr) reactants were synthesized in powder form. Tin-based perovskites ($\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$)) were deposited as a thin film on glass substrates using the ultrasonic spray pyrolysis (USP) method. X-ray diffraction (XRD) was used to examine the crystallographic characteristics of the synthesized MAI/MABr powders and perovskite thin films. A shift occurred in the XRD peaks by changing the I/Br ratios. Morphological analysis of the MAI and MABr were carried out by scanning electron microscopy (SEM). While the average particle size was calculated a $\sim 94 \mu\text{m}$ for MAI, it was obtained as $\sim 188 \mu\text{m}$ for MABr. The Fourier transform infrared (FTIR) spectroscopy peaks observed for synthesized MAI and MABr were found to be compatible with commercial MAI and MABr FTIR peaks. Elemental analysis of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films was performed energy dispersive spectroscopy (EDS). Forbidden band gap (E_g) values of perovskite thin films were obtained from Tauc curves. The E_g value increased with an increasing I/Br ratio. The deposition of highly stoichiometric $\text{MASnI}_x\text{Br}_{3-x}$ perovskites thin films was achieved by the USP method. This method has many parameters need to be optimized. This study gives optimum parameters that are difficult to determine.

Keywords Ultrasonic spray pyrolysis · Advanced energy materials · Perovskite · Band gap engineering

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1 Introduction

Perovskite materials with exceptional semiconductor capabilities that result from the combination of organic and inorganic components in the ABX_3 (where A is an organic (CH_3NH_3^+ , MA, or $\text{HC}(\text{NH}_2)_2^+$, FA) or alkali metal (Cs^+) cation, B is a divalent metal (Pb, Sn or Ge) and X is a halide anion) perovskite structure are referred to as hybrid perovskite materials [1, 2]. These perovskites are preferred in electronic and optoelectronic devices with their desirable properties, such as high carrier mobility, long carrier diffusion length, tunable band gap, low exciton binding energy, and high absorption coefficient [3]. The stability problem of Pb halide perovskites and the toxicity of Pb are disadvantages for solar cell applications [4]. Therefore, Sn or Ge, which are in the same group in the periodic table, can be used as an alternative to Pb [5]. However, the Sn^{2+} in MASnI_3 perovskites is oxidized to Sn^{4+} , rapidly. This case is limited their stability [6]. To provide the stability, the use of $(\text{I}_{1-x}\text{Br}_x)$ instead of only I in the halide part was tried and it was observed that the stability increased without losing efficiency (12.3%) [7].



With power conversion efficiency increasing from 3.8 to 26% in solar cell applications, interest in perovskite materials is increasing [8, 9]. Perovskite materials are used as a light-absorbing layer in solar cells applications. Band gap engineering, which involves changing the material composition to provide the appropriate band gap for perovskite materials to absorb light, can be used to tune the forbidden band gap of the perovskite layer [10, 11]. By varying the cations, metals, or halides used for the perovskite material, the band gap can be adjusted in the range of about 1.2–3.0 eV [7, 12, 13]. Low-cost solution-based methods can be used to prepare perovskite materials. To create perovskite thin films, reactants can be dissolved in appropriate solutions and coated on surfaces using various thin film coating methods [14]. Examples to these methods are spin coating, dip coating, blade coating, spray pyrolysis, and ultrasonic spray pyrolysis (USP) [15, 16]. The USP method is a thin-film coating technique that enables uniform films on large areas. Thin films with high surface quality can be obtained by controlling parameters such as substrate temperature, shaping gas pressure, solution flow rate, and nozzle-substrate distance with the USP method [16–18]. It has been reported that ~ 80% material savings are achieved in the formation of thin films with the USP method compared to other coating methods [19].

The studies on perovskite solar cells generally focus on their stability problems and the improvement of efficiency [20, 21]. The third branch of the studies is the scalable production of PSCs, which is important for commercialization [22]. While several efforts on large-area deposition of Pb-based perovskite thin films are available in the literature, there is a lack of information about Sn-based perovskites. In our study, the deposition of $\text{MASnI}_x\text{Br}_{3-x}$ perovskite thin films was carried out by the ultrasonic spray pyrolysis method, which is a suitable method for the large-area fabrication of PSCs. However, this method has many parameters, such as nozzle-substrate distance, substrate temperature, and perovskite ink concentration that need to be optimized. The aim of our study is to present optimized parameters that require a lot of effort to determine.

In this study, MAI and MABr were synthesized in powder form. Fourier transform infrared (FTIR) spectroscopy was used to compare the molecular bond structures of the synthesized and commercial MAI and MABr, while X-ray diffraction (XRD) was used to examine their crystallographic structures. $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite solutions were obtained, and then deposited as a thin film on glass substrates using the USP method. Structural and elemental analyzes of these thin films were performed by XRD and energy dispersive spectroscopy (EDS), respectively. The forbidden band gap (E_g) values of perovskite thin films were calculated using the Tauc plot. A comparison in the form of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) was found in only two studies

in the literature [23, 24]. However, $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskites were not produced by the USP method in these studies. The presented study is the first to compare the structural, morphological, and optical properties of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskites produced by the USP method.

2 Experimental Processes

While methylamine (CH_3NH_2 , 40% in methanol, TCI) and hydrogen iodide (HI, 57% in H_2O , Sigma) were used for the synthesis of methylammonium iodide ($\text{CH}_3\text{NH}_3\text{I}$, (MAI)), CH_3NH_2 , and hydrogen bromide (HBr, 47% in H_2O , Merck) were used for the synthesis of methylammonium bromide ($\text{CH}_3\text{NH}_3\text{Br}$, (MABr)). Methylamine was taken into separate 100 mL flasks and cooled at 0 °C. HI/HBr was added dropwise to methylamine in an ice bath with the help of a dropping funnel. Reactions were stirred in an ice bath at 0 °C in a controlled manner for 2 h. At the end of the reaction, the solvents in the flasks were evaporated from the environment at 190 mbar in the rotary evaporator set at 50 °C. The white solid powders MAI and MABr formed in the flasks were washed three times with diethyl ether ($(\text{C}_2\text{H}_5)_2\text{O}$) with the help of a Buchner funnel. Thus, the impurities were partially removed. MAI and MABr powders were dried in a muffle furnace at 60 °C for 2 days. MAI and MABr were treated with ethanol/diethyl ether (1:3) to remove impurities. The reactants dissolved in ethanol/ethyl acetate (1:3) solution were then placed in the refrigerator overnight to crystallize. MAI and MABr powders dried in a muffle furnace at 60 °C for one night were stored in a glovebox. Previous works inspired the synthesis of MAI and MABr reactants [25, 26]. Images of synthesized MAI and MABr were given in Fig. 1. The reactants used to prepare perovskite solutions and the solution preparation parameters are given in Table 1.

SnI_2 (TCI), SnBr_2 (Sigma), and *n, n*-dimethylformamide (DMF, Merck) were commercially available. Perovskite solutions were prepared by mixing them in a glove box at 80 °C for 45 min. Glass substrates were cut in dimensions of $2.0 \times 2.0 \times 0.1$ cm. Substrates were ultrasonically cleaned using detergent, ethanol, acetone, and ethanol-double-distilled water. The substrates were then dried with nitrogen gas. Perovskite solutions were coated on glass substrates with an ultrasonic spray pyrolysis (USP) device (Sono-Tek Corporation, 2012 Route 9 W) using the coating parameters given in Table 1. The deposition process is shown schematically in Fig. 2.

XRD measurements of MAI and MABr powders and $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskites deposited on glass substrates by the USP method were taken using Cu-K α radiation ($\lambda = 1.5418$ Å) of Bruker D8 Advance Twin-Twin diffractometer (40 kV, 40 mA). The analysis of these

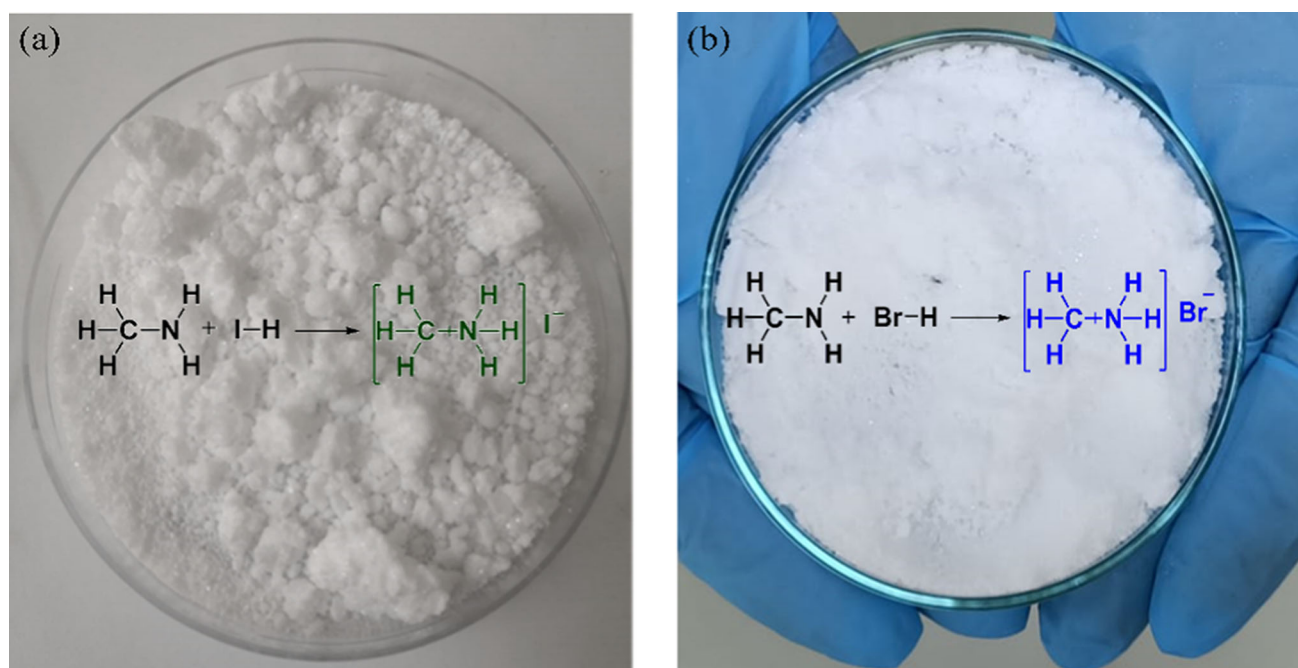


Fig. 1 Images of synthesized a MAI b MABr

Table 1 Solution preparation and USP coating parameters for $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films

Parameters	Perovskites			
	MASnI ₃	MASnI ₂ Br	MASnIBr ₂	MASnBr ₃
Substrate temperature (°C)	150	180	180	180
Substrate- nozzle distance (cm)	7.0	7.0	7.0	7.0
Nozzle frequency (kHz)	120	120	120	120
Flow rate (mL/dk)	0.10	0.15	0.15	0.15
Oxygen ratio (< %)	1.5	1.5	1.5	1.5
Shaping nitrogen gas pressure (%99,999) (psi)	1.0	1.0	1.0	1.0
Nozzle cooling (psi)	15	15	15	15
Engine cooling (psi)	2.0	2.0	2.0	2.0
Power (W)	5.33 W	5.33 W	5.33 W	5.33 W
MAI–MABr–SnI ₂ –SnBr ₂ Molarity	0.10 M–0.07 M	0.09 M–0.09 M	0.10 M–0.10 M	0.11 M–0.11 M
MAI–MABr–SnI ₂ –SnBr ₂ mass (g)	0.066–0.104	0.038–0.130	0.061–0.107	0.043–0.105
DMF volume (mL)	4.0	4.0	4.0	3.5

measurements was made with the help of the same company's EVA software and ICDD database. Fourier transform infrared (FTIR) measurements of MAI and MABr were taken with FTIR spectrophotometer (JASCO FT/IR 4700). Surface morphologies and elemental analyzes of perovskite films were performed with scanning electron microscopy (SEM) (FEI Quanta FEG 250) added to the EDAX-EDS system. Optical absorption studies of the films were taken with an ultraviolet–visible region (UV–Vis) spectrophotometer (PG-Instruments T80 +).

3 Results and Discussion

XRD measurements for MAI and MABr were taken in the range of $5^\circ \leq 2\theta \leq 65^\circ$. XRD patterns of MAI and MABr are given in Fig. 3a and b, respectively. The XRD data obtained for commercial (green line) and synthesized (blue line) of both MAI and MABr are in agreement. The XRD pattern of the synthesized MAI is in consistency with the PDF 00-010-0737 card (red bar chart in Fig. 3a) while the XRD pattern of the synthesized MABr is in consistency with the PDF 00-010-0699 card (red bar chart in Fig. 3b). For synthesized MAI and



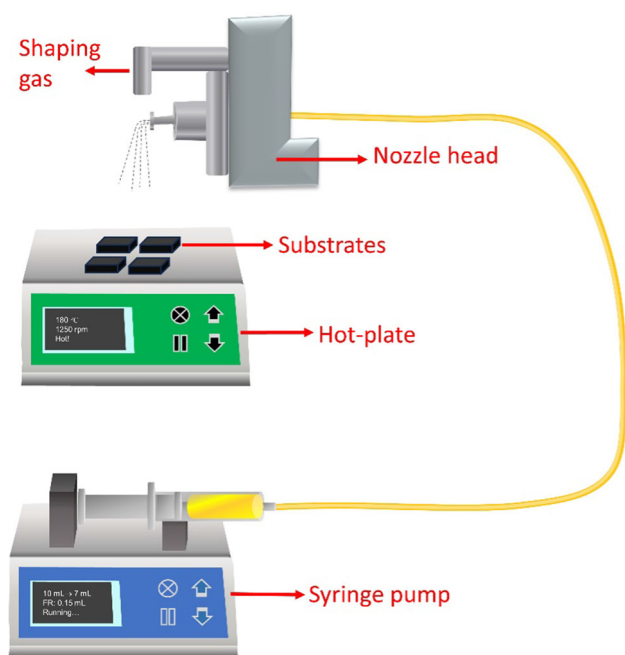


Fig. 2 Schematic representation of the deposition process

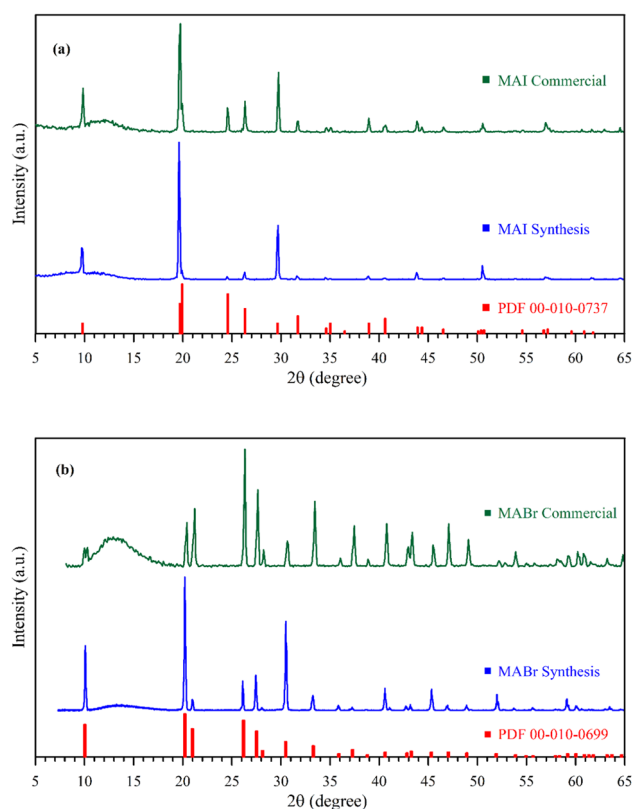


Fig. 3 XRD patterns for both commercial and synthesized **a** MAI **b** MABr

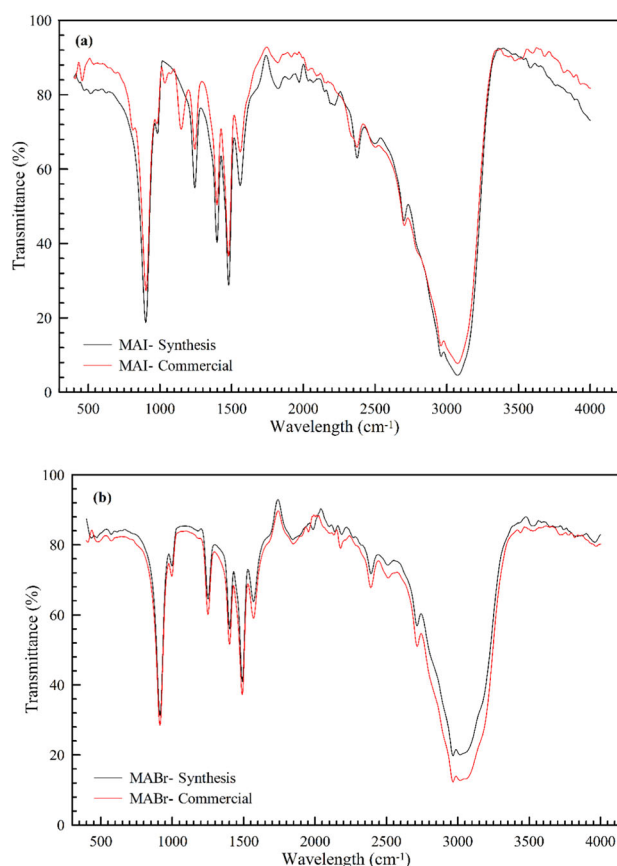


Fig. 4 FTIR spectra for both commercial and synthesized **a** MAI **b** MABr

MABr, the peak intensity is higher at $\sim 20^\circ$. This peak at $\sim 20^\circ$ can be considered as the main peak, as it exhibits higher intensity than the others. It has been reported in previous studies that this peak shows the orientation for MAI (101), while it shows (002) for MABr [27, 28]. It is seen that the synthesized MAI and MABr peaks do not exhibit an extra peak compared to the peaks in commercial and PDF cards (Fig. 3a and b). However, the peak intensities of $\sim 20^\circ$ and $\sim 21^\circ$ for commercial MABr powder appear to differ from the peaks indicated in the synthesis and PDF card. Thus, it can be said that the synthesized MABr has a better crystal structure than the commercial MABr. This is an indication that both compounds are free of impurities and can be synthesized in pure form. Purification with ethanol/diethyl is also thought to be effective in obtaining MAI and MABr crystals free from contamination.

FTIR spectra for synthesized and commercial MAI and MABr reactants are given in Fig. 4a and b, respectively. C–H stretch peaks (902 , 1402 , 1490 , 2965 cm^{-1}), N–H peaks (1251 , 1402 , 1569 , 3016 cm^{-1}), C–N (997 cm^{-1}), and CN–NH (2715 cm^{-1}) bands for MAI were detected in Fig. 4a. C–H stretch peaks (914 , 1398 , 1479 , 2960 cm^{-1}), N–H peaks (1243 , 1398 , 1560 , 3075 cm^{-1}), C–N

(982 cm^{-1}), and CN–NH bands (2701 cm^{-1}) of MABr were determined from Fig. 4b. The peaks obtained for each mode were found to be in agreement with those previously reported [29, 30]. The compatibility of functional groups with the literature and commercial MAI and MABr peaks for the synthesized powder reactants is an indication that a pure crystal structure has been obtained.

MAI and MABr organic halide salts are used in the optoelectronics industry as precursors for the preparation of perovskite materials used in many electronic applications, such as photovoltaic cells, perovskite solar cells, and light-emitting diodes (LEDs) [31–33]. The fact that these materials can be easily synthesized provides convenience in terms of both cost and time savings. It is a critical point to synthesize a pure and original crystal structure. It has been reported in the literature that the problem to be overcome in perovskites is stability [34, 35].

It is known that perovskites with large grain size quality crystallization are tolerant of crystal defects and have less chance of degradation [36]. MAI and MABr precursors are useful for synthesizing mixed cation or anion perovskites, which are essential for improving the carrier diffusion length, band gap, and power conversion efficiency of perovskite solar cells. Therefore, it is important that the precursor materials used in perovskite materials have a quality crystal structure. Tan et al. reported that methylammonium (MA^+) cations can form perovskites with higher crystallinity. It has also been reported that MA^+ cations can heal deep trap zones due to their large radius and dipolar structure [37]. Synthesis MAI and MABr organic salts have been used in the production of many perovskites in the literature [23, 27, 28, 32, 38]. XRD and FTIR analyses showed successful synthesis of MAI and MABr reactants in this study. So, synthesis MAI and MABr reactants were used to produce tin-based hybrid perovskite thin films by using the USP method.

SEM images for the synthesized MAI and MABr are given in Fig. 5a and b, respectively. The particle sizes of reactants are also plotted on SEM images. It was observed that the particle sizes for MAI ranged from 30 to 160 μm while the particle sizes for MABr ranged from 150 to 260 μm . It is seen from the SEM images that agglomerations are formed and the reactants have a particulate structure. The average particle size for MAI was $\sim 94 \mu\text{m}$ while the average particle size for MABr was calculated as $\sim 188 \mu\text{m}$. Leupold et al. reported that the defect density decreased as the MAI reactant size increased. The researchers defined MAI crystals ranging in size from 25 to 1300 μm as large MAI crystals and obtained an average MAI crystal size of 213 μm [39]. In this study, it was understood that the synthesis MAI and MABr reactants have grain sizes that can be used in perovskites.

XRD patterns of MASnI_3 perovskite deposited on glass substrates using synthesized MAI and commercial SnI_2 are

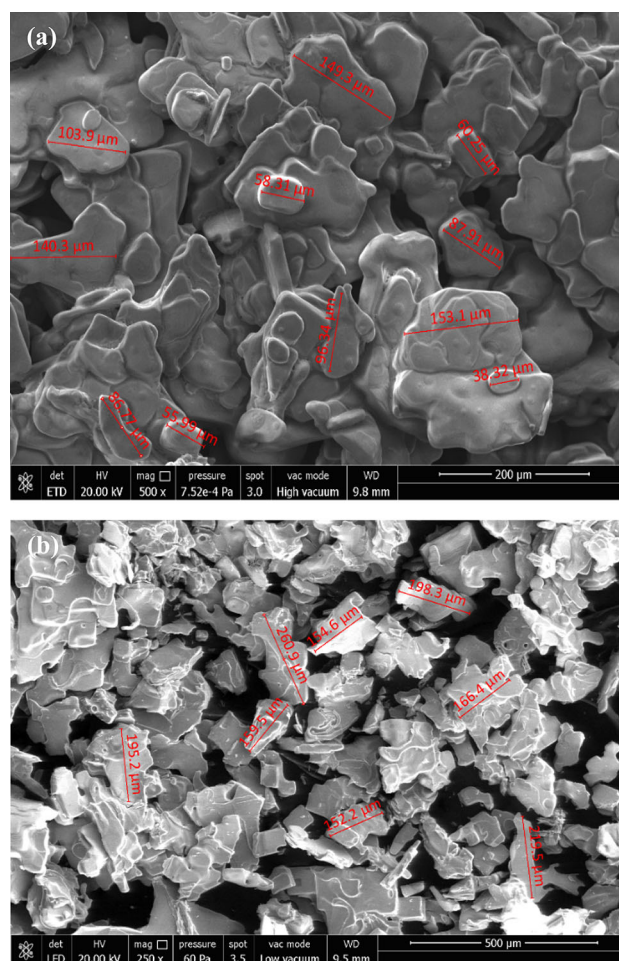


Fig. 5 SEM images for synthesized **a** MAI **b** MABr

shown in the range of $10^\circ \leq 2\theta \leq 80^\circ$ (Fig. 6). XRD measurements were taken immediately after MASnI_3 coating under laboratory conditions (23 $^\circ\text{C}$, 10–15% relative humidity). It was observed that MASnI_3 perovskite thin films have a tetragonal crystal structure. It has a peak in the (100) plane at $\sim 14^\circ$ and a peak in the (220) plane at $\sim 28^\circ$. Furthermore, the MASnI_3 perovskite thin films crystallographic structure appears to be consistent with the PDF 00-080-4670 card. XRD patterns of MASnI_2Br (green line), MASnIBr_2 (blue line), and MASnBr_3 (red line) perovskite thin films deposited on glass substrates are shown in the range of $10^\circ \leq 2\theta \leq 80^\circ$ (Fig. 7).

These perovskite structures were prepared using synthesized MAI and MABr reactants and commercial $\text{SnI}_2/\text{SnBr}_2$ ionic salts. The positions of the peaks in the given XRD patterns are in agreement with the peak positions reported in the literature [23, 24]. XRD measurements revealed that $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films were effectively deposited on glass substrates using the USP method.



Fig. 6 XRD patterns for MASnI_3 perovskite thin films deposited on glass substrates by the USP method

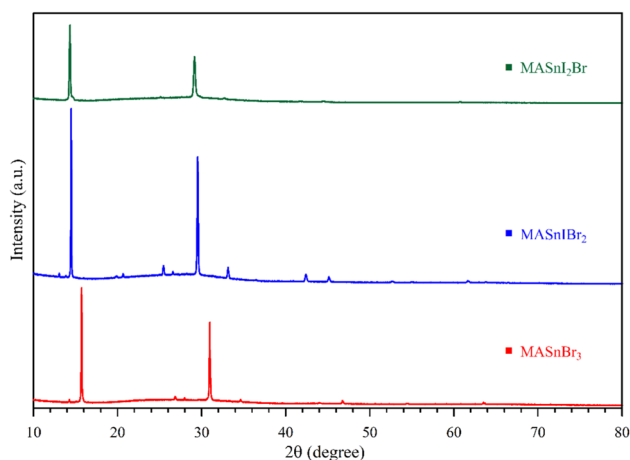
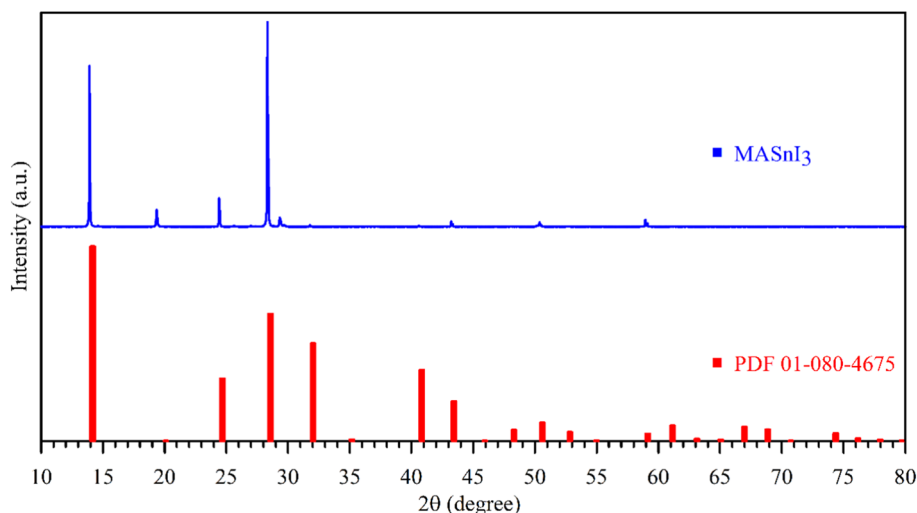


Fig. 7 XRD patterns of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 2, 1, 0$) perovskites deposited on glass substrates by the USP method

The peaks around 15° (100) and around 30° (200) are the characteristic peaks of the $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 2, 1, 0$) perovskite structure. Hao et al., and Hsu et al., included the calculated and experimentally obtained XRD patterns of $\text{MASnI}_x\text{Br}_{3-x}$ perovskite structures in their studies [23, 40]. The intensity of the (100) peak in the calculated XRD patterns in these studies is higher than the intensity of the (200) peak. This is in agreement with our experimental work. Also, as can be seen in Fig. 7, there is a leftward shift in the 2θ axis of the peaks with the increase in the I/Br ratio. The change in the lattice constant as a result of the change in the I/Br ratio is the cause of the shift in the peaks [23]. EDS analyses for MASnI_3 , MASnI_2Br , MASnIBr_2 , and MASnBr_3 perovskite thin films were shown in Fig. 8a–d, respectively. The composition of these perovskite thin films is given in the inset of the figures. EDS analysis indicated that the appropriate stoichiometric ratios for $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films were attained (Fig. 8a–d).

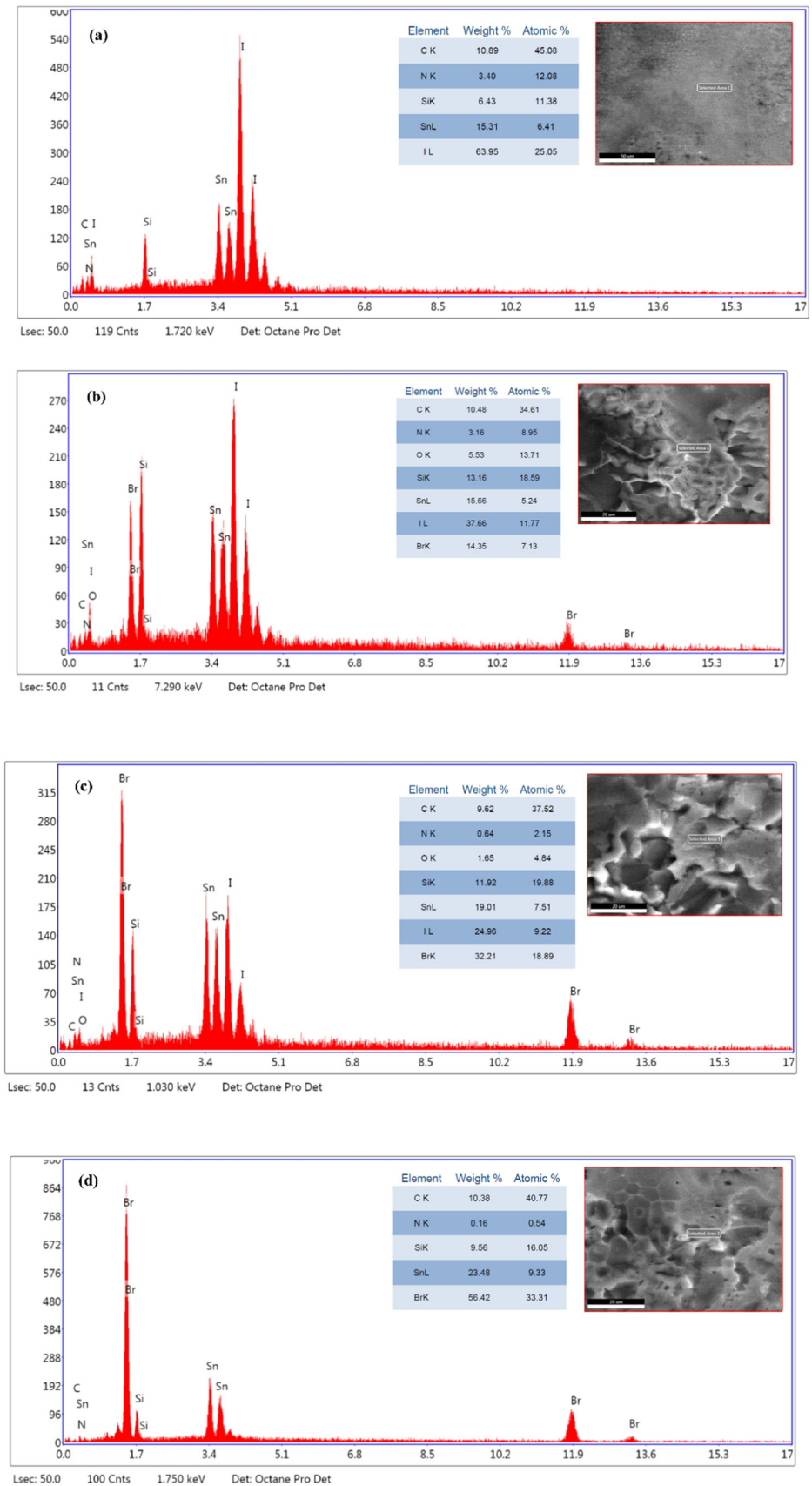
The band gap of a semiconductor (E_g) can be determined by the Tauc curve. UV–Vis measurements are used to obtain the Tauc curves. The E_g can be found from the point where the Tauc curve, obtained by plotting $(\alpha h\nu)^2$ with respect to $h\nu$, intersects the x -axis. Here, α is the absorption coefficient, h is Planck's constant, and ν is the frequency of the incident photon [41]. The Tauc curves of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films are given in Fig. 9. E_g values for MASnBr_3 , $\text{MASnI}_2\text{Br}_2$, MASnI_2Br , and MASnI_3 perovskite thin films were calculated as 2.11 eV, 1.71 eV, 1.51 eV, and 1.47 eV, respectively. Some E_g values reported in the literature for $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 2, 1, 0$) perovskite structures are summarized in Table 2.

As can be seen from Table 2, the E_g values calculated for $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 2, 1, 0$) perovskite structures are compatible with previously studies [23, 24]. However, the E_g value for MASnI_3 was higher than the value reported in the literature [23, 24, 42–44]. This may be due to the humidity sensitivity of MASnI_3 perovskite. During UV–Vis measurements, MASnI_3 perovskite is likely to decompose into MAI and Sn^{+4} upon exposure to air humidity. This may cause the band gap to increase [45]. It was observed that E_g increased as the amount of Br increased in $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films. The increase in the E_g value shows that the band gap can be controlled by changing the halogen ion.

4 Conclusion

Powdered MAI and MABr reactants were synthesized. XRD, FTIR, and SEM analyses revealed that these reactants were successfully obtained. It was observed that tin-based perovskite thin films ($\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$)) could be deposited uniformly on glass substrates using the USP

Fig. 8 EDS analysis of **a** MASnI_3 **b** MASnI_2Br **c** MASnIBr_2 **d** MASnBr_3 perovskite thin films



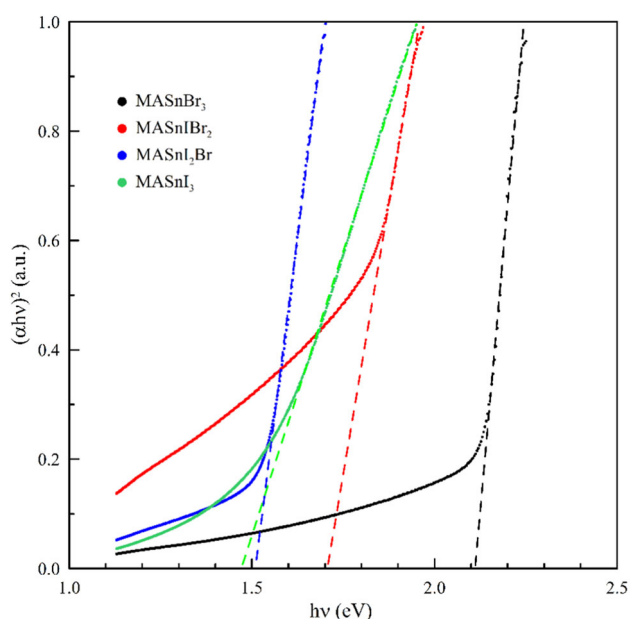


Fig. 9 The Tauc curves of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) perovskite thin films

Table 2 Some E_g values reported in the literature for $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 2, 1, 0$) perovskite thin films

Perovskites	Fabrication method	E_g (eV)	References
MASnI_3	Spin-coating	1.23	[42]
		1.30	[23]
	Spin-coating with SnF_2 additive	1.25	[43]
	Spin-coating one step	1.35	[44]
	Vapor-assisted solution process	1.27	
	Low-temperature vapor-assisted solution process	1.26	
MASnI_2Br		1.25	[24]
MASnI_2Br	Spin-coating	1.56	[23]
MASnIBr_2		1.75	
MASnBr_3		2.15	

method. The peak at $\sim 30^\circ$ for the MASnBr_3 perovskite structure became visible in the XRD pattern as the Br ratio increased. As the I/Br ratio changed, a change was observed in the XRD peaks. This is likely to be produced by a change in perovskite's lattice constant. It was seen from the EDS analysis that stoichiometric $\text{MASnI}_x\text{Br}_{3-x}$ perovskite thin films were deposited by the USP method. This shows that the USP coating parameters are optimized to offer the best performance in the production of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2, 1, 0$) thin films. The E_g values of $\text{MASnI}_x\text{Br}_{3-x}$ ($x = 3, 2,$

1, 0) perovskite thin films increased as the Br content in perovskite increased. In the preparation of the halide part of the perovskite, the manipulation was performed by applying the $\text{I}_x\text{Br}_{3-x}$ formulation instead of I_3 . This study will be a guide for scalable deposition of Sn-based perovskite solar cells.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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